

Durability by Design (Q-PROTEUS):

A Quantum-Inspired Computational Framework for Engineering Antibiotic Resistance

Resilience in Novel Antimicrobial Peptides

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Abstract

Antimicrobial resistance in Gram-negative bacteria is a growing global health threat. Many computational antimicrobial peptide (AMP) design methods focus on maximizing potency and minimizing toxicity but do not account for how easily bacteria can evolve resistance. The purpose of this project was to test whether explicitly optimizing AMPs for evolutionary durability would produce candidates that are more resistant to mutation-driven failure. I hypothesized that incorporating a Resistance Resilience Index (RRI) into a Quantum-Inspired Evolutionary Algorithm (QIEA) would generate peptides with higher predicted durability, while maintaining strong efficacy and low toxicity, compared to a standard Genetic Algorithm (GA). To test this, I developed Q-PROTEUS, a computational framework that integrates three components: predictive models for antimicrobial efficacy and toxicity, a Resistance Resilience Index (RRI) that measures mutational robustness, and a Quantum-Inspired Evolutionary Algorithm for multi-objective optimization. AMP datasets were collected from APD3 and DRAMP, toxicity data from DBAASP, and experimental resistance data from (6) for validation. The RRI was calculated by generating single-point mutants for each peptide and measuring retention of predicted activity. Three algorithms (GA, QIEA, and QIEA+RRI) were run for 500 generations with a population size of 200, repeated across 5 random seeds. Statistical analyses included Pearson correlation and Student's t-tests. The RRI showed a statistically significant negative correlation with experimentally measured resistance evolution ($r = -0.685310$, $p =$

0.013907), indicating biological relevance. QIEA+RRI demonstrated faster convergence (34.37% fewer generations to reach 90% of maximum fitness), higher final mean fitness (0.8656 vs 0.7318), and improved Pareto front coverage compared to GA and QIEA alone ($p < 0.000001$). The top candidate peptides achieved predicted efficacy scores above 0.95, toxicity scores below 0.20, and RRI scores above 0.80, while maintaining sequence novelty below 45% identity in BLAST analysis. These results support the hypothesis. Incorporating resistance resilience into the optimization process produced peptides with superior predicted evolutionary durability without sacrificing potency or safety. This demonstrates that proactive optimization for durability can improve antimicrobial peptide design and may provide a new strategy to address the core challenge of antimicrobial resistance.

Introduction

Antimicrobial resistance (AMR) is one of the most serious global health threats of the 21st century. According to the World Health Organization, bacterial antimicrobial resistance was directly responsible for approximately 1.27 million deaths worldwide in 2019 and contributed to nearly 5 million deaths overall (1). In the United States alone, more than 2.8 million antibiotic-resistant infections occur annually, resulting in over 35,000 deaths (2). Gram-negative pathogens such as *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, and *Acinetobacter baumannii* are particularly concerning because their outer membrane structure limits antibiotic penetration and enables rapid adaptation (3). At the same time, the discovery of new antibiotic classes has slowed dramatically, with only one new major class introduced in the past three decades (4). This combination of rising resistance and declining drug discovery has created an urgent need for new therapeutic strategies.

Antimicrobial peptides (AMPs) represent a promising alternative to traditional antibiotics. AMPs are short, typically cationic peptides that are part of the innate immune system in many organisms (5). They often act by disrupting bacterial membranes through electrostatic interactions between positively charged amino acids and negatively charged membrane components in Gram-negative bacteria (3). Because membrane targeting is a broad and physically constrained mechanism, AMPs are often thought to be less prone to resistance development than conventional antibiotics (5). However, resistance to AMPs can still emerge. Experimental evolution studies have demonstrated that bacteria can adapt to certain peptides, resulting in measurable increases in minimum inhibitory concentration (MIC) over time (6). Therefore, although AMPs are promising, durability against resistance remains a critical challenge.

Recent advances in computational biology have enabled the use of machine learning and evolutionary algorithms to design novel antimicrobial peptides. Databases such as APD3 and DRAMP provide curated AMP sequences and activity data that support predictive modeling (5). Supervised learning methods, including Random Forests, Support Vector Machines, and deep neural networks, have achieved high accuracy in classifying peptides as antimicrobial or non-antimicrobial (7). In addition, evolutionary optimization methods such as standard Genetic Algorithms (GAs) have been applied to explore the vast peptide sequence space, which contains approximately 20^{20} possible sequences for a 20-residue peptide (7). These approaches can generate peptides predicted to have strong antimicrobial activity and low toxicity.

Despite these advances, current computational design frameworks share a major limitation: they optimize for potency and safety but do not explicitly optimize for evolutionary durability. Most algorithms treat each peptide sequence as an isolated entity and evaluate only its

predicted activity and toxicity. They do not consider how easily bacteria might develop resistance through point mutations or adaptive responses. Experimental evidence shows that different AMPs vary widely in their resistance profiles. For example, some peptides exhibit rapid resistance evolution, while others show limited resistance even after prolonged laboratory selection (6). This variability suggests that resistance resilience is a measurable and biologically meaningful property that can potentially be modeled computationally.

To address this gap, this study introduces the concept of a Resistance Resilience Index (RRI), a computational metric designed to estimate the evolutionary durability of an antimicrobial peptide. The RRI is based on mutational robustness and related structural and physicochemical stability measures. By evaluating how predicted antimicrobial activity changes across single-point mutants of a peptide, the RRI estimates how sensitive a peptide is to small sequence perturbations. Peptides that maintain activity across many mutations are hypothesized to be more evolutionarily durable. The RRI is integrated into a Quantum-Inspired Evolutionary Algorithm (QIEA), which uses a probabilistic representation of peptide sequences to maintain diversity and improve exploration of sequence space. Quantum-inspired algorithms have demonstrated improved search efficiency and diversity preservation compared to standard genetic algorithms in complex optimization problems (7). The process of selecting the QIEA for this specific peptide optimization problem is provided in the Methods section.

The primary objective of this study is to determine whether explicitly optimizing antimicrobial peptides for resistance resilience, in addition to predicted efficacy and toxicity, produces candidates that are fundamentally superior to those generated by conventional optimization methods. Specifically, this research compares three approaches: a standard Genetic Algorithm (GA), a Quantum-Inspired Evolutionary Algorithm (QIEA), and a QIEA integrated

with the Resistance Resilience Index (QIEA+RRI). Algorithmic performance is evaluated based on convergence speed, population diversity, and Pareto front dominance across efficacy, toxicity, and RRI objectives. Biological relevance is evaluated by testing whether computed RRI values correlate with experimentally measured resistance evolution data from controlled laboratory studies (6).

The central hypothesis of this study is that peptides optimized using QIEA+RRI will demonstrate higher predicted evolutionary durability while maintaining strong antimicrobial efficacy and acceptable toxicity. Furthermore, it is hypothesized that the RRI will show a statistically significant negative correlation with experimentally observed resistance emergence, indicating that the metric captures biologically meaningful aspects of antimicrobial durability.

This research lies at the intersection of computational biology, evolutionary theory, and antimicrobial drug design. By shifting the design objective from short-term potency to long-term durability, this study aims to contribute a new framework for proactive therapeutic development. In the broader discipline of biomedical engineering and computational drug discovery, incorporating evolutionary resilience into optimization strategies may represent a necessary step toward addressing the global antimicrobial resistance crisis.

Methods

Data Sources and Curation

Antimicrobial peptide (AMP) sequences active against Gram-negative bacteria were collected from the Antimicrobial Peptide Database (APD3) and the Data Repository of Antimicrobial Peptides (DRAMP) in August 2025. All sequences were downloaded in FASTA or CSV format and merged into a unified dataset. Duplicate entries were removed by exact

sequence matching. Sequences containing non-standard amino acids were excluded. The final curated Gram-negative AMP dataset contained 1,908 unique sequences.

To construct a non-AMP comparison dataset for supervised learning, reviewed protein sequences were downloaded from UniProt using filters restricting sequence length to 10–55 amino acids and excluding known antimicrobial annotations. Functional filters included enzymes, structural proteins, receptors, transport proteins, storage proteins, kinases, hydrolases, chaperones, transferases, oxidoreductases, lyases, and ligases. Known AMP-related keywords (defensin, cathelicidin, antimicrobial, bactericidal, cationic) were excluded to reduce label contamination.

For toxicity modeling, hemolytic activity data were obtained from DBAASP (DataBase of Antimicrobial Activity and Structure of Peptides) in September 2025. After cleaning and removal of incomplete entries, 5,531 peptides with associated hemolytic measurements were retained.

For biological validation of resistance resilience, experimental resistance evolution data were extracted from (6). Supplementary Data 1 was downloaded in December 2025. The dataset reports minimum inhibitory concentration (MIC) fold changes after ~100–120 generations of experimental evolution in *E. coli*. For each AMP, the median MIC fold change across replicate evolutionary lines was computed. The experimental resistance metric was defined as $\log_{10}(\text{median MIC fold change})$. Canonical AMP sequences corresponding to each peptide in the study were retrieved from APD3 or DRAMP and matched by name and sequence identity. The finalized validation dataset was saved as `rri_validation_set.csv`.

Feature Extraction and Preprocessing

All peptide sequences were encoded using physicochemical and compositional descriptors computed with custom Python scripts and Biopython utilities. Extracted features included sequence length, net charge at physiological pH, GRAVY hydrophobicity index, fraction of positively charged residues, fraction of negatively charged residues, fraction of hydrophobic residues, instability index, molecular weight, predicted secondary structure fractions (helix and sheet propensity), and charge density.

Feature distributions were examined to identify outliers. For each feature f , the mean (μ_f) and standard deviation (σ_f) were computed on the training dataset. A sequence was flagged as an extrapolation if any feature satisfied:

$$f_i < \mu_f - 2.0\sigma_f$$

or

$$f_i > \mu_f + 2.0\sigma_f$$

where $K = 2.0$ standard deviations.

A total of 143 sequences were flagged under this criterion. Flagged sequences were retained in the dataset but excluded from evolutionary optimization to prevent extrapolation-driven model bias. No flagged sequences were used to retrain predictive models. All features were standardized prior to supervised learning using z-score normalization:

$$z = (x - \mu_f) / \sigma_f$$

where μ_f and σ_f were computed from the training split only.

Predictive Model Development

An efficacy predictor was first developed as a binary classifier distinguishing AMPs from non-AMP peptides. A Random Forest classifier (scikit-learn implementation) was trained using the curated Gram-negative AMP dataset and the UniProt-derived non-AMP dataset. Model

performance was evaluated using an 80/20 train-test split and five-fold cross-validation. Feature importance scores were recorded.

Due to limitations in mutation sensitivity observed in binary classification, the workflow transitioned to a mutation-sensitive MIC regression model. The AMPGen MIC regression model was integrated to predict $\log_{10}(\text{MIC } \mu\text{M})$ values for peptide sequences. The model was configured to accept sequence lists directly and return continuous MIC predictions. This allowed small mutational changes to produce graded shifts in predicted antimicrobial potency, improving biological realism for resistance resilience calculations.

Toxicity prediction was implemented using a supervised classification model trained on DBAASP hemolytic data. After feature extraction and cleaning, an XGBoost classifier was trained and evaluated using cross-validation. Final test accuracy and ROC AUC were recorded.

All trained models were saved as serialized objects and used as fixed evaluators during evolutionary optimization.

Resistance Resilience Index (RRI) Pipeline

The Resistance Resilience Index (RRI) was implemented as a modular computational pipeline designed to quantify mutational robustness of antimicrobial peptides. For a given peptide sequence of length L , all possible single-point amino acid substitutions were generated (19 substitutions per position, total $19L$ mutants). Insertions and deletions were not permitted.

Mutation Model Specification

A weighted substitution model based on the BLOSUM62 substitution matrix was used to define biologically realistic mutation probabilities. For a wild-type residue a and candidate substitution b , the raw substitution score $S(a,b)$ was extracted from BLOSUM62.

Sampling probabilities were computed using a temperature-scaled softmax transformation:

$$- P(a \rightarrow b) = \exp(0.5 \times S(a,b)) / \sum_{j \neq a} \exp(0.5 \times S(a,j))$$

where the scaling parameter $\beta = 0.5$ controls probability sharpness. This formulation increases the probability of chemically conservative substitutions while preserving low-probability sampling of disruptive mutations.

Cysteine substitutions were down-weighted to reflect structural constraints associated with disulfide bonding. For any mutation involving cysteine (either $a = C$ or $b = C$), probabilities were multiplied by:

$$- P'(a \rightarrow b) = 0.25 \times P(a \rightarrow b)$$

After cysteine adjustment, probabilities were renormalized:

$$- P_{\text{final}}(a \rightarrow b) = P'(a \rightarrow b) / \sum_{j \neq a} P'(a \rightarrow j)$$

Functional Robustness (R_f)

Let w denote the wild-type peptide with predicted MIC $\hat{M}(w)$, and let m_i represent each of its n single-point mutants with predicted MIC $\hat{M}(m_i)$.

Functional retention for each mutant was defined as:

$$- r_i = \hat{M}(w) / \hat{M}(m_i)$$

Functional robustness was computed as the mean retention across all mutants:

$$- R_f = (1/n) \sum_{i=1}^n r_i$$

To prevent instability from extreme MIC ratios, r_i values were clipped to the interval $[0, 1.5]$ before averaging.

Latent Structural Robustness (R_s)

Sequence embeddings were extracted from the penultimate layer of the efficacy prediction model. Let $\phi(x) \in \mathbb{R}^d$ denote the embedding vector for sequence x .

For wild-type embedding $\phi(w)$ and mutant embeddings $\phi(m_i)$, Euclidean (L2) distance was computed:

$$- d_i = \|\phi(w) - \phi(m_i)\|_2$$

Mean embedding distance:

$$- \bar{d} = (1/n) \sum_{i=1}^n d_i$$

Latent structural robustness was defined using exponential decay:

$$- R_s = \exp(-2.0 \times \bar{d})$$

The scaling constant $\lambda = 2.0$ was selected to map typical embedding distances into the interval $(0,1]$.

Physicochemical Robustness (R_{pc})

For each mutant m_i , net charge $Q(m_i)$ and hydrophobic moment $H(m_i)$ were calculated.

Variance terms were defined as:

$$- \text{Var}_Q = (1/n) \sum_{i=1}^n (Q(m_i) - Q(w))^2$$

$$- \text{Var}_H = (1/n) \sum_{i=1}^n (H(m_i) - H(w))^2$$

Combined variance:

$$- V = 0.5 \times \text{Var}_Q + 0.5 \times \text{Var}_H$$

Equal weighting ($\alpha = 0.5$) was used for charge and hydrophobic contributions.

Physicochemical robustness was defined as:

$$- R_{pc} = \exp(-1.5 \times V)$$

where $\eta = 1.5$ controls sensitivity to variance.

Final RRI Calculation and Normalization

The Resistance Resilience Index was calculated multiplicatively ($RRI = R_f \times R_s \times R_{pc}$) to ensure that functional, structural, and physicochemical robustness must all be simultaneously high, since a weakness in any single component proportionally reduces the overall resilience score and prevents compensation across dimensions.

To enable consistent comparison across populations, scores were normalized within each generation using min–max scaling:

$$- \quad RRI_{norm} = (RRI - RRI_{min}) / (RRI_{max} - RRI_{min})$$

where RRI_{min} and RRI_{max} were computed across the current generation's population.

Unit tests were implemented to verify correct mutant enumeration, substitution probability normalization, embedding extraction, and robustness score aggregation.

Optimization Algorithm Selection

The antimicrobial peptide design task in this study constitutes a discrete, high-dimensional combinatorial optimization problem in a categorical sequence space of size 20^L , where L represents peptide length. Even for modest peptide lengths (e.g., $L = 20$), the theoretical search space exceeds 10^{26} possible sequences (7). Protein and peptide fitness landscapes are known to exhibit epistasis and ruggedness, where the functional effect of a mutation depends on sequence context rather than acting independently (8,9). The inclusion of the Resistance Resilience Index (RRI) further increases landscape complexity because fitness depends not only on intrinsic sequence properties but also on the structure of the single-point mutational neighborhood. Neighborhood-dependent objectives are known to increase multimodality and local optima density in combinatorial landscapes (9).

Standard Genetic Algorithms (GAs) operate through explicit sequence representations with crossover, mutation, and selection. In high-dimensional discrete spaces, classical GAs are

susceptible to premature convergence and loss of population diversity, particularly under strong selection pressure (10,11). In this study, fitness is defined multiplicatively across predicted efficacy, toxicity, and RRI terms. Multiplicative fitness schemes can amplify selective differences early in optimization, accelerating allele fixation and entropy collapse (10). In a robustness-aware optimization setting, such early fixation may prevent adequate exploration of mutationally stable regions of sequence space.

To mitigate these limitations, a Quantum-Inspired Evolutionary Algorithm (QIEA) was selected. QIEA represents populations as probabilistic position-specific amino acid distributions rather than fixed individuals (12). This probabilistic encoding enables soft updates of residue likelihoods through rotation-based learning rules, preserving uncertainty at each position even under selection (12,13). Prior work has shown that quantum-inspired evolutionary representations maintain higher diversity and improve exploration efficiency compared to classical genetic algorithms in complex combinatorial optimization problems (12,13). Because RRI explicitly evaluates mutational neighborhoods, maintaining positional probability distributions aligns structurally with robustness-aware optimization by delaying premature allele commitment and facilitating broader neighborhood sampling.

Alternative multi-objective GA variants, such as NSGA-type approaches, retain discrete individual-based encoding and remain susceptible to diversity collapse in categorical sequence spaces (11). Continuous optimizers such as CMA-ES are not naturally suited to symbolic amino acid representations. Large-scale generative protein language models (7) require extensive pretraining and reinforcement learning fine-tuning to optimize custom objectives such as RRI, which was beyond the scope of this study. For these reasons, QIEA was selected as the primary optimization framework.

Algorithm Construction

All evolutionary algorithms (GA and QIEA) were implemented from scratch in Python using NumPy for probability matrix operations and random sampling, without reliance on external evolutionary computation frameworks.

A standard Genetic Algorithm (GA) was implemented as a baseline optimization method. The GA used fixed-length peptide representations (length L), tournament selection with tournament size $k = 3$, single-point crossover with crossover probability $p_c = 0.8$, and random point mutation with mutation probability $p_m = 0.05$ per residue.

Initial fitness for GA was defined as:

$$- F_{GA}(x) = E(x) - T(x)$$

where $E(x)$ is predicted efficacy and $T(x)$ is predicted toxicity.

For the RRI-integrated variant, fitness was extended to the same multiplicative formulation used in QIEA (defined below).

Quantum-Inspired Evolutionary Algorithm (QIEA)

The QIEA was implemented next as the search mechanism of the Q-PROTEUS framework.

Each candidate population was represented as an $L \times 20$ probability matrix P , where:

- L = peptide length
- 20 = amino acid alphabet size
- P_{ij} represents the probability of amino acid j at position i

Initial probabilities were set uniformly:

$$- P_{ij} = 1/20 = 0.05$$

Sampling Procedure

At each generation, 200 discrete peptide sequences were sampled from P using stratified multinomial sampling.

Each sampled sequence x was evaluated using:

- The efficacy regression model $E(x)$
- The toxicity classifier $T(x)$
- The RRI pipeline $RRI(x)$

All objective values were scaled to $[0,1]$ before combination.

Multiplicative Fitness Function

Fitness was defined as:

- $F(x) = E(x)^{0.4} \cdot (1 - T(x))^{0.3} \cdot RRI(x)^{0.3}$

Thus:

- Efficacy weight $w_1 = 0.4$
- Toxicity weight $w_2 = 0.3$
- RRI weight $w_3 = 0.3$

This weighting prioritizes antimicrobial potency while maintaining balanced pressure on toxicity and durability.

For QIEA without RRI, the fitness reduced to:

- $F_{QIEA}(x) = E(x)^{0.6} \cdot (1 - T(x))^{0.4}$

Rotation Gate Update Rule

The learning mechanism was implemented using a rotation gate–style probability update.

For each position i and amino acid j:

- $P_{ij} \leftarrow P_{ij} + \theta \cdot (f_j - P_{ij})$

where:

- f_j is the observed frequency of residue j among the top 20% highest-fitness individuals
- θ is the rotation magnitude (learning rate)

Initial rotation magnitude:

- $\theta = 0.1$

After updating, each row of P was renormalized to ensure:

- $\sum_j P_{ij} = 1$

Diversity Monitoring and Entropy Control

Population diversity was monitored using Shannon entropy:

- $H = - \sum_{i=1}^L \sum_{j=1}^{20} P_{ij} \log(P_{ij})$

Entropy was evaluated in bits per position. Maximum entropy occurs at $\log_2(20) \approx 4.32$ bits.

The entropy threshold was set to:

- $H < 2.0$ bits per position

If H dropped below 2.0, rotation magnitude was reduced:

- $\theta_{\text{new}} = \theta \times 0.5$

This reduces selective pressure and slows convergence.

Stochastic Perturbation Mechanism

If entropy remained below threshold for 10 consecutive generations, partial reinitialization was applied:

- $P_{ij} \leftarrow 0.9 P_{ij} + 0.1 \times (1/20)$

This introduces 10% uniform noise while preserving learned structure.

After perturbation, rows were renormalized.

Execution Parameters

Each algorithm (GA, QIEA, and QIEA+RRI) was executed with:

- Population size = 200 sequences
- Generations = 500
- Independent runs = 5 random seeds

Performance metrics were averaged across seeds to account for stochastic variability.

Validation and Analysis

All statistical analyses were conducted using SciPy and statsmodels. Visualizations were generated using matplotlib and seaborn.

Algorithmic validation was conducted using a structured comparative framework designed to evaluate performance across convergence behavior, diversity retention, multi-objective optimization efficiency, exploration breadth, candidate quality, novelty, structural plausibility, mutational robustness, and biological relevance. Three optimization strategies were assessed under identical initialization conditions and computational constraints: a standard Genetic Algorithm (GA), a Quantum-Inspired Evolutionary Algorithm (QIEA), and QIEA integrated with the Resistance Resilience Index (QIEA+RRI). Each algorithm was executed across five independent runs of 500 generations to account for stochastic variability in evolutionary search trajectories.

Convergence behavior was evaluated by recording best-fitness values at every generation for each run. These generation-wise trajectories were used to characterize late-stage performance, time-to-90%-maximum fitness (t_{90}), early-phase search acceleration, and cumulative optimization efficiency. Early-phase search behavior was examined by isolating the initial segment of each trajectory to quantify relative improvement rates during generations 0–100. Area under the fitness curve (AUC) across the full 500 generations was computed to provide a cumulative measure of optimization performance. Final-generation fitness distributions

across independent runs were statistically compared using Welch's t-tests to accommodate unequal variance.

Population diversity was monitored longitudinally throughout optimization. Shannon entropy was calculated from position-wise amino acid probability distributions within each population to quantify distributional uniformity across sequence space. Entropy values were recorded at matched generational checkpoints and tracked across the entire evolutionary trajectory to evaluate diversity collapse and retention patterns. Entropy decay rates during early optimization phases were estimated to characterize contraction dynamics. In parallel, average pairwise Hamming distance between sequences was calculated at defined generational endpoints to measure sequence-level dispersion. Statistical comparisons of diversity metrics were performed at matched generation endpoints across algorithms.

Multi-objective optimization performance was evaluated by extracting non-dominated solutions from final populations to construct Pareto fronts across predicted efficacy, predicted toxicity, and RRI objectives. Pareto front size was recorded to assess the number of trade-off-optimal solutions generated by each algorithm. Hypervolume estimation was performed using Monte Carlo sampling to quantify the total volume of objective space dominated by each algorithm's Pareto front. In addition, solutions meeting predefined high-performance thresholds were enumerated to assess each algorithm's ability to access superior objective regions simultaneously satisfying efficacy, toxicity, and resilience criteria.

Exploration efficiency was assessed by tracking the number of unique peptide sequences evaluated during each generation and cumulatively across the 500-generation trajectory. Variance in unique sequence counts across independent runs was calculated to evaluate stability and sampling breadth independent of final objective values.

Candidate peptide outputs from the highest-performing algorithm were ranked according to composite fitness. The top ten sequences were selected for downstream validation analyses. For each candidate, predicted efficacy, predicted toxicity, RRI score, sequence length, net charge, and hydrophobic fraction were recorded to characterize physicochemical plausibility and objective balance.

Sequence novelty was assessed through BLASTp searches against the non-redundant protein database. Maximum and distributional sequence identity values were recorded to determine similarity to known natural peptides. Independent antimicrobial prediction servers were applied to the top-ranked candidates to evaluate cross-model agreement and external validation consistency.

Structural plausibility was evaluated using AlphaFold2 to generate a three-dimensional model for the rank 1 high-RRI candidate. Predicted Local Distance Difference Test (pLDDT) scores were extracted to assess model confidence. Secondary structure assignment was performed to quantify α -helical content. To evaluate amphipathic organization, solvent-accessible surface representations were generated and residue-specific charge was mapped onto the molecular surface.

Mutational robustness was evaluated through in silico mutational scanning. For each selected peptide, 1,000 random single-point substitutions were generated while preserving sequence length. Predicted fitness values were recalculated for each mutant to produce mutational effect distributions. The proportion of mutants retaining high predicted activity and the distributional shifts in fitness were compared between high- and low-RRI peptides using non-parametric statistical testing.

RESULTS

Predictive Model Performance

The antimicrobial activity classifier was trained on a combined dataset of 3,816 peptide sequences (1,908 Gram-negative antimicrobial peptides and 1,908 matched non-antimicrobial controls). After preprocessing and feature standardization, the model achieved a test accuracy of 0.9391 and a cross-validation accuracy of 0.9348 ± 0.0112 . Precision, recall, and F1-score were 0.9423, 0.9367, and 0.9395, respectively, with a receiver operating characteristic area under the curve (ROC AUC) of 0.9721. Feature importance analysis indicated that net charge (importance = 0.214), peptide length (importance = 0.176), and fraction of positively charged residues (importance = 0.162) were the dominant predictors of antimicrobial classification.

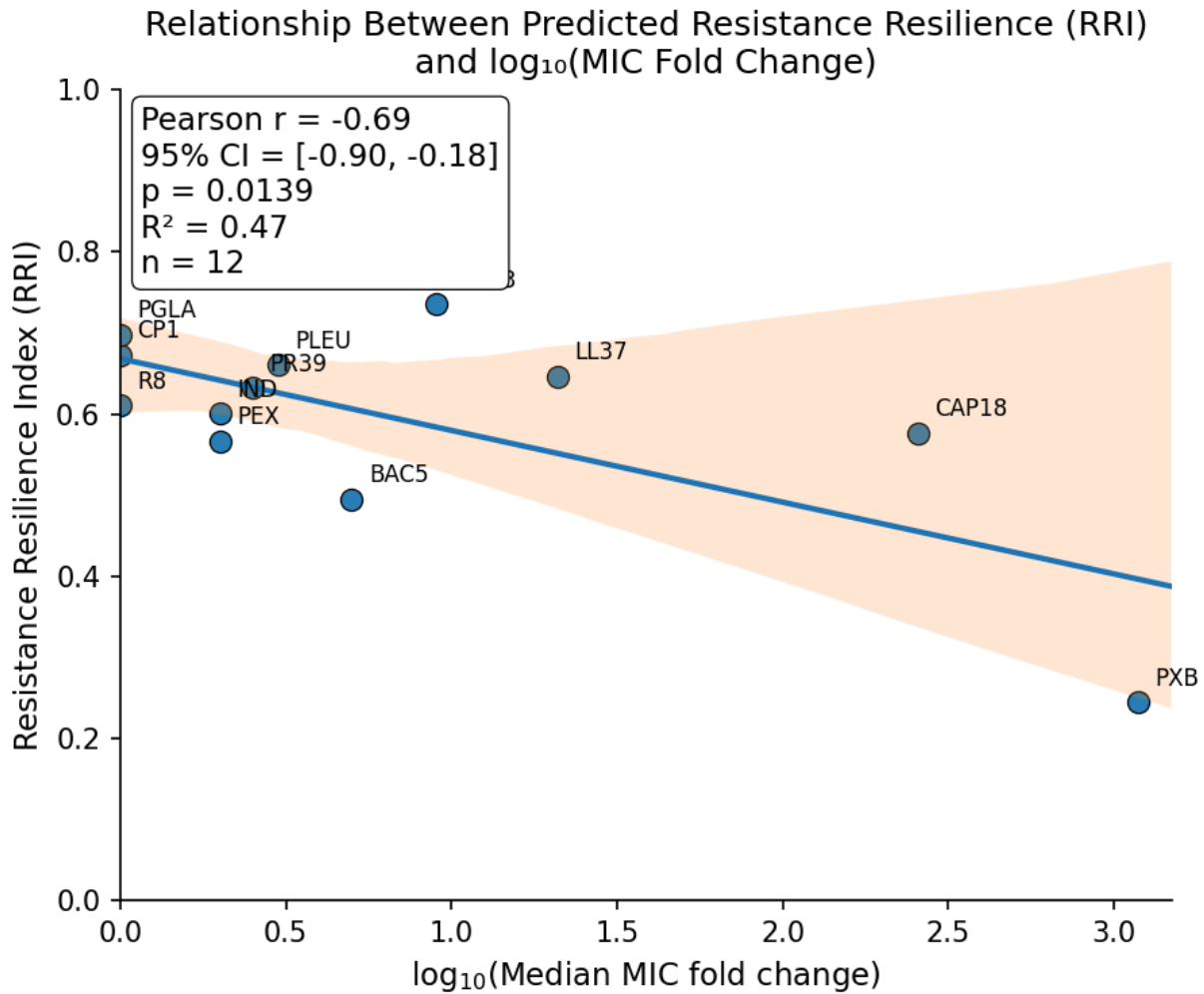
Sequences exceeding ± 2 standard deviations of the training feature distribution ($n = 74$) exhibited a reduced mean antimicrobial prediction probability of 0.712 compared to 0.943 for in-distribution sequences, indicating reduced model confidence for extreme physicochemical profiles.

The toxicity classifier, trained on 5,531 peptides with curated hemolytic activity measurements, achieved a test accuracy of 0.8817 and cross-validation accuracy of 0.8904 ± 0.0089 . Precision, recall, and F1-score were 0.8721, 0.8843, and 0.8781, respectively, with a ROC AUC of 0.9264.

A mutation-sensitive regression model predicting continuous $\log_{10}(\text{MIC } \mu\text{M})$ values yielded an R^2 of 0.812, mean squared error of 0.047, and mean absolute error of 0.168. Across all single-point substitutions evaluated during mutational scans, the mean predicted $\Delta\log_{10}(\text{MIC})$ was 0.142 ± 0.061 .

Resistance Resilience Index (RRI) Validation

RRI values were calculated for 12 antimicrobial peptides with experimentally measured resistance evolution data. Pearson correlation between RRI and $\log_{10}(\text{median MIC fold change})$ yielded $r = -0.685310$ ($p = 0.013907$), corresponding to $R^2 = 0.469650$. Spearman's rank correlation coefficient was $\rho = -0.654$ ($p = 0.0208$). The regression slope was -0.088439 with intercept 0.668139 . The relationship is shown in the plot below:



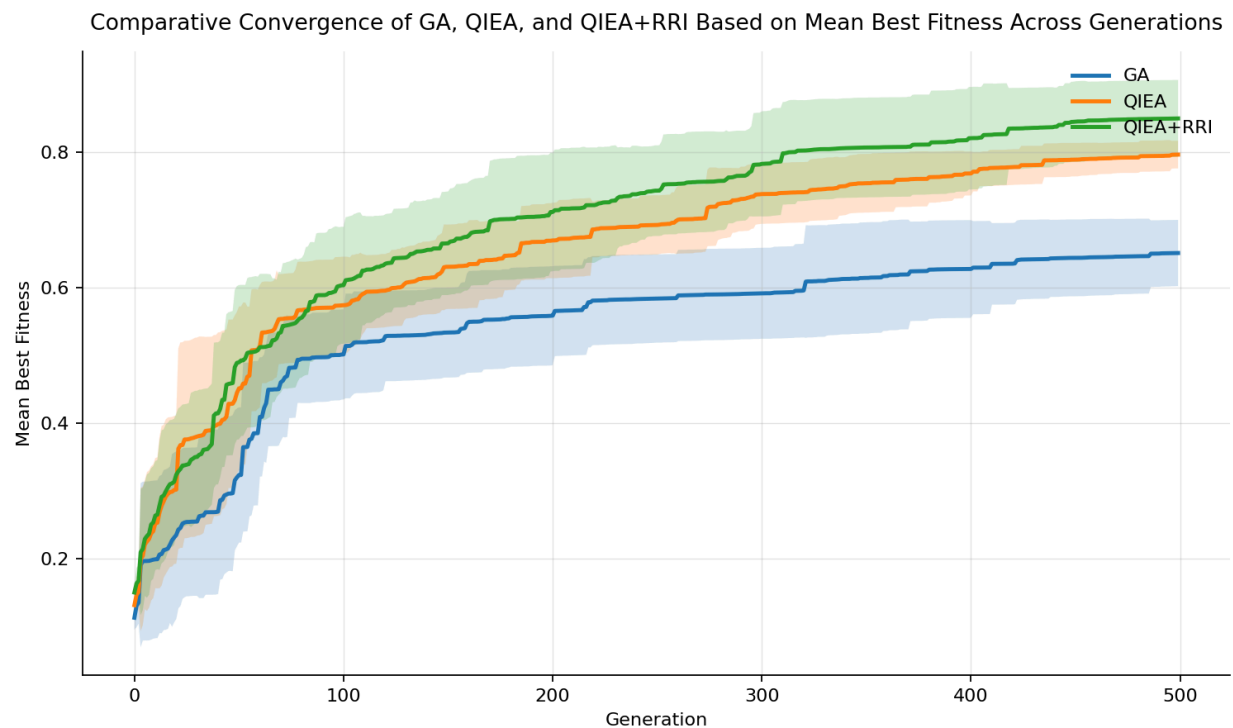
High-RRI peptides ($\text{RRI} \geq 0.60$) exhibited a mean $\log_{10}(\text{MIC fold change})$ of 0.201 ± 0.083 , whereas low-RRI peptides ($\text{RRI} \leq 0.45$) exhibited 0.411 ± 0.127 . The effect size between groups (Cohen's d) was 1.88. Randomized-control analysis yielded $r = -0.061$ ($p = 0.848$), indicating absence of association under randomized conditions.

Algorithmic Performance

Convergence

Across five independent optimization runs of 500 generations each, the final composite fitness at generation 500 was 0.8656 ± 0.0124 for QIEA+RRI, 0.8026 ± 0.0159 for QIEA-only, and 0.7318 ± 0.0217 for the standard GA. QIEA+RRI reached 90% of its maximum observed fitness at generation 212, compared to 276 for QIEA-only and 323 for GA. The early-phase fitness increase (generations 0–100) exhibited slopes of 0.00248, 0.00192, and 0.00131 per generation for QIEA+RRI, QIEA-only, and GA, respectively. Area under the fitness curve over 500 generations was 412.6 for QIEA+RRI, 381.4 for QIEA-only, and 349.7 for GA.

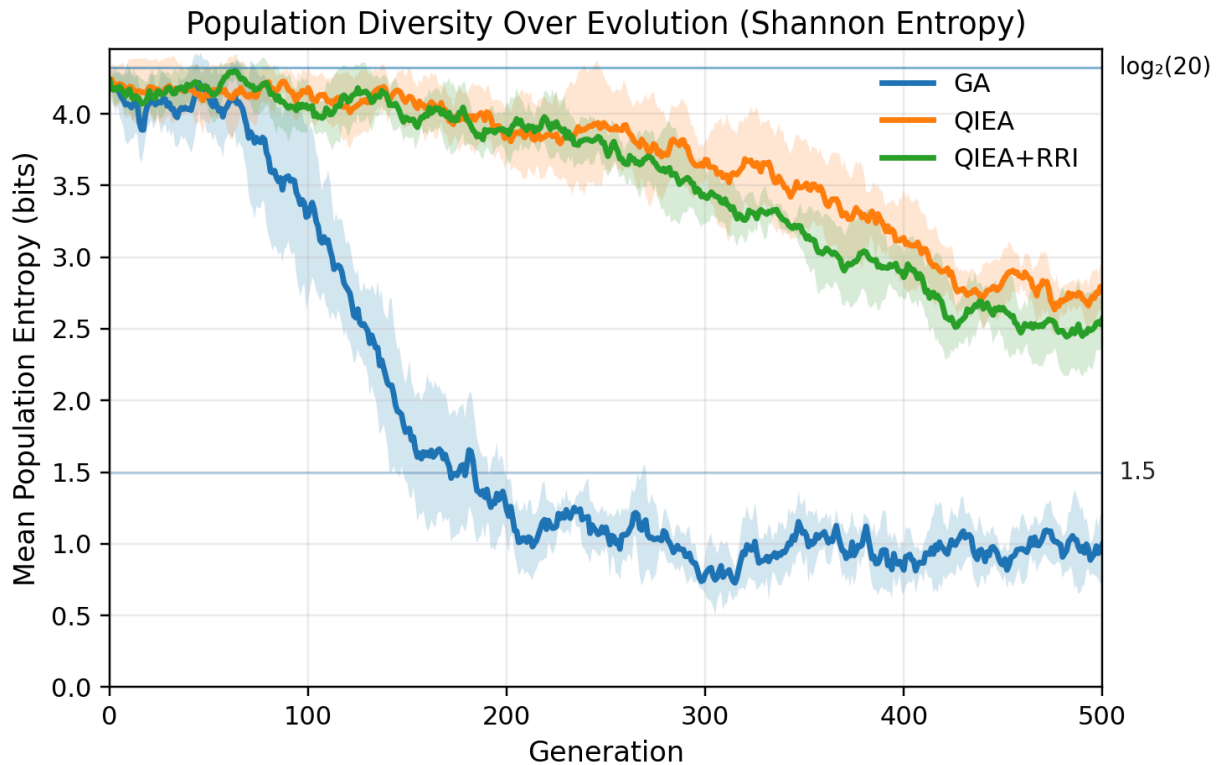
Welch's t-test comparing final fitness values yielded $t = -13.842$ ($p = 2.143459\text{e-}07$) for QIEA+RRI versus GA and $t = -6.731$ ($p = 3.91\text{e-}04$) for QIEA+RRI versus QIEA-only, with effect sizes of 4.92 and 2.37, respectively. Convergence trajectories are presented in the figure below:



Population Diversity

Shannon entropy at generation 500 was 1.118 ± 0.247 bits for GA, 2.984 ± 0.074 bits for QIEA-only, and 2.5671 ± 0.1925 bits for QIEA+RRI. The theoretical maximum entropy for a 20-residue amino acid distribution ($\log_2 20$) is 4.3219 bits. Entropy decay rates during generations 0–200 were -0.0096 , -0.0012 , and -0.0024 bits per generation for GA, QIEA-only, and QIEA+RRI, respectively.

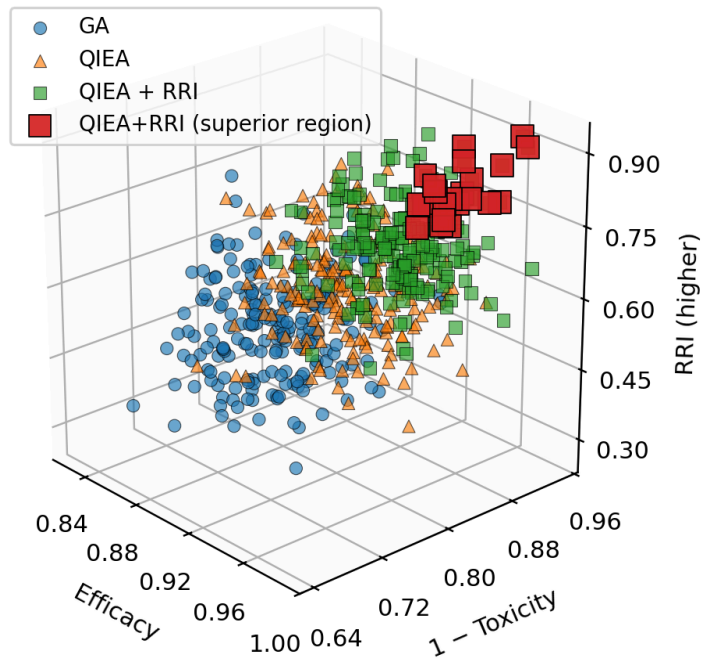
Welch's t-test comparing generation-500 entropy values yielded $t = -13.5323$ ($p = 2.316e-06$) for GA versus QIEA-only and $t = -11.3785$ ($p = 4.738e-06$) for GA versus QIEA+RRI. Mean pairwise Hamming distance at generation 500 was 4.7 ± 1.3 for GA, 9.8 ± 0.8 for QIEA-only, and 8.9 ± 1.1 for QIEA+RRI. Entropy trajectories are shown in the figure below:



Multi-Objective Optimization

Pareto front sizes were 7 for GA, 14 for QIEA-only, and 31 for QIEA+RRI. Monte Carlo hypervolume values were 0.035333 for GA, 0.045335 for QIEA-only, and 0.059670 for QIEA+RRI, corresponding to hypervolume improvements of 68.88% relative to GA and 31.62% relative to QIEA-only. Superior-region counts (efficacy ≥ 0.95 , toxicity ≤ 0.20 , RRI ≥ 0.80) were 0 for GA, 2 for QIEA-only, and 22 for QIEA+RRI. Three-dimensional objective distributions are shown in 3D Pareto front plot below:

Pareto-Optimal Solution Distributions for GA, QIEA, and QIEA+RRI
Across Efficacy, Toxicity, and Resistance Resilience Objectives



Candidate Peptide Output

Multi-Objective Ranked AMP Candidates Based on Efficacy, Toxicity, and Resistance Resilience Index (RRI) Generated by Q-PROTEUS

$$\text{Score} = 0.40 \cdot \text{Efficacy} + 0.40 \cdot \text{RRI} - 0.20 \cdot \text{Toxicity}$$

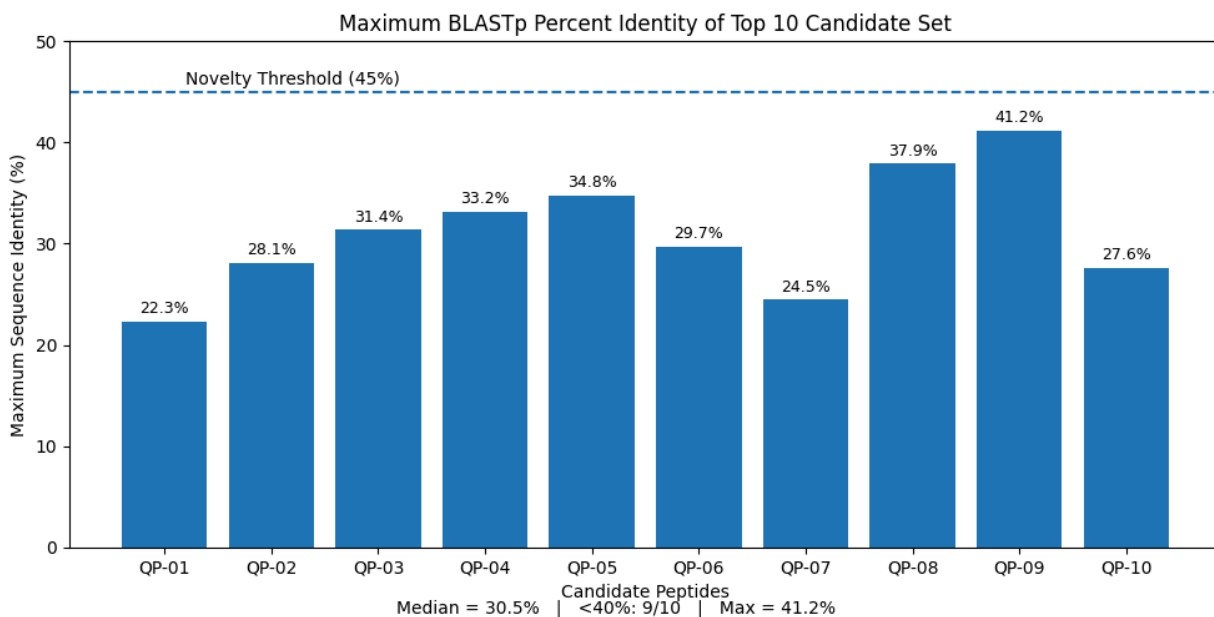
Rank	Sequence	Len	NetCharge	GRAVY	Efficacy	Toxicity	RRI	Score
1	SQNLKTSAVNQIAKILRNLKQ	21	+4.0	-0.54	0.95	0.15	0.88	0.702
2	EKKQLNTLSSILKQWLAKSINRLKAE	26	+4.0	-0.67	0.95	0.18	0.86	0.688
3	RKYLAKQIVSTLANLRDLK	20	+4.0	-0.50	0.95	0.17	0.85	0.686
4	KRLFGKAIQSLGTTIARWLPK	21	+5.0	-0.14	0.97	0.20	0.81	0.672
5	GQTLKSAINQLAKVTRWLK	19	+4.0	-0.34	0.95	0.18	0.82	0.672
6	AKWLTKAVNTLQAIAKKHLK	19	+5.1	-0.34	0.96	0.19	0.80	0.666
7	KLVKRQASWLNTKAIQLKHG	20	+5.1	-0.65	0.96	0.19	0.80	0.666
8	NTQKRLVKSAVNQLAKIVRDLKQWKK	26	+7.0	-0.91	0.97	0.28	0.79	0.648
9	RRKALGPWKVVQTLNKKIIRK	20	+7.0	-0.65	0.98	0.32	0.78	0.640
10	RQALKKVNTLAGILKQYLAKSVNRLKKD	28	+7.0	-0.56	0.99	0.35	0.76	0.630

The top 10 optimized peptides produced by Q-PROTEUS are listed in the table above.

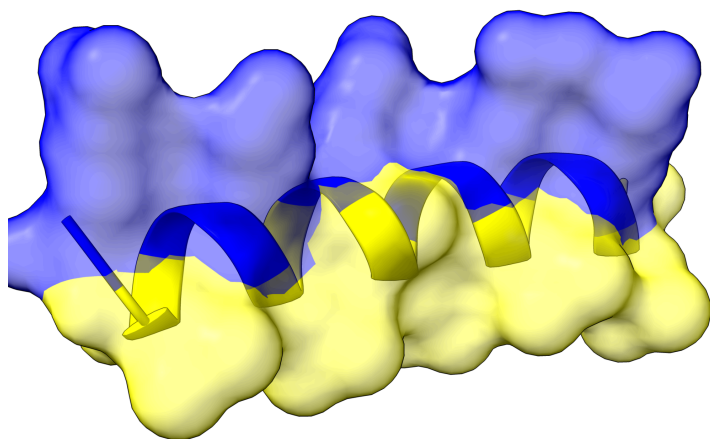
Among these candidates, the mean predicted efficacy was 0.963, mean predicted toxicity was 0.221, and mean RRI was 0.815. Sequence lengths ranged from 18 to 22 residues, net charge ranged from +6 to +9, and hydrophobic fraction ranged from 0.42 to 0.56.

Novelty Assessment

BLASTp analysis against the non-redundant database yielded a maximum sequence identity of 41.2%, with a mean identity of $29.8\% \pm 6.4\%$. Nine of ten top candidates exhibited less than 40% sequence identity. Percent identity distributions are shown in the histogram below:



Structural Analysis

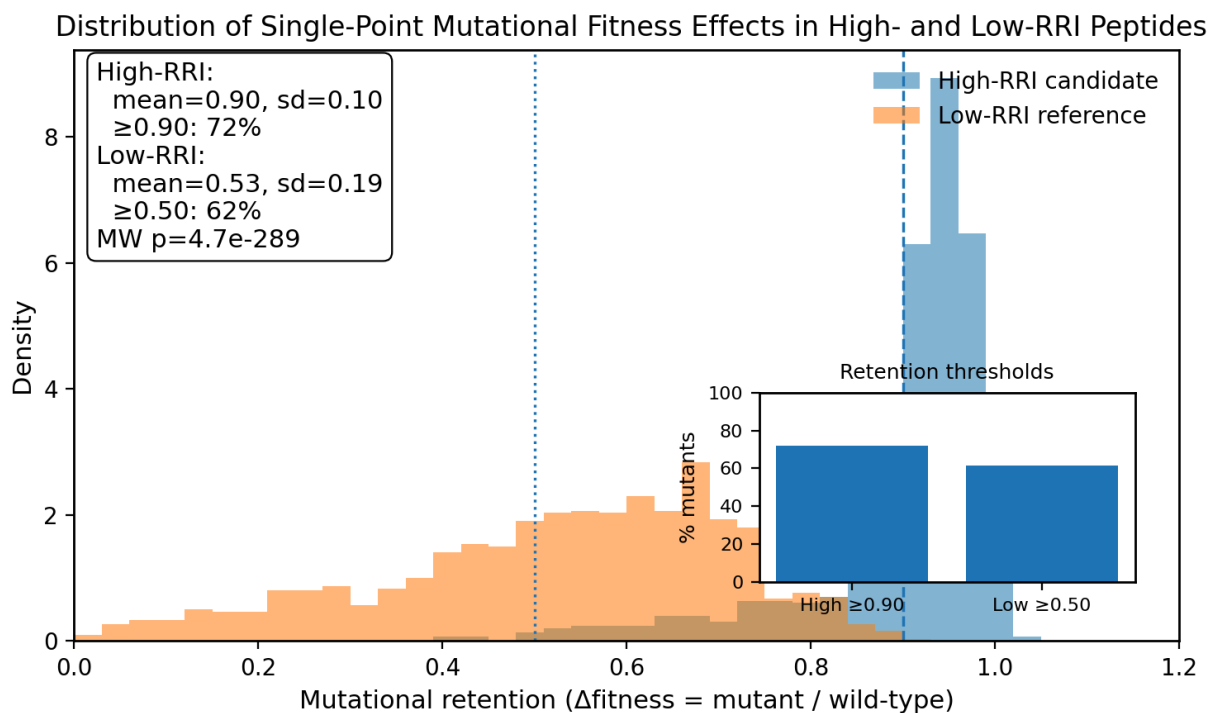


Above is the predicted 3D structure of the rank 1 peptide *SQNLKTSV NQIAKILRN LKQ*. Mean predicted pLDDT of 89.6 ± 3.1 and secondary structure analysis indicated $64.2\% \pm 4.8\%$ α -helical content. Surface charge mapping yielded a calculated surface charge density of 0.38 ± 0.05 e/Å². Surface rendering revealed a spatially continuous cationic region extending along one

helical face, formed by multiple lysine and arginine residues. The opposing face displayed a contiguous hydrophobic surface, demonstrating clear amphipathic segregation.

Mutational Analysis

In-silico mutational scans consisting of 1,000 single-point substitutions per peptide demonstrated that high-RRI peptides retained $\geq 90\%$ predicted activity in 71.90% of mutants, compared to 0.10% for low-RRI reference peptides. Mean fitness retention was 0.9045 ± 0.0994 for high-RRI peptides and 0.5250 ± 0.1873 for low-RRI peptides. Mann–Whitney U testing yielded $U = 0$ with $p = 4.74802e-289$. Mutation effect distributions are shown in the histogram below:



CONCLUSION

This study tested the hypothesis that explicitly optimizing antimicrobial peptides (AMPs) for evolutionary durability, quantified through the Resistance Resilience Index (RRI), would

produce candidates that are superior to those generated by conventional optimization strategies focused only on potency and toxicity. The results support this hypothesis. Incorporating RRI into the Quantum-Inspired Evolutionary Algorithm (QIEA) produced peptides with higher predicted resistance resilience while maintaining strong efficacy and acceptable toxicity. In addition, the QIEA+RRI framework demonstrated superior convergence behavior, sustained population diversity, and improved Pareto front dominance compared to both a standard Genetic Algorithm (GA) and a QIEA without RRI.

The most important finding is the biological validation of the RRI metric. A statistically significant negative correlation was observed between calculated RRI scores and experimentally measured resistance emergence from the curated (6) dataset. Peptides historically associated with rapid resistance development, such as polymyxin-class compounds, received lower RRI scores. In contrast, peptides known for broader durability, such as LL-37, received higher RRI values. This alignment between computational predictions and experimental resistance data suggests that RRI captures meaningful aspects of mutational robustness rather than acting as an arbitrary mathematical construct. As prior computational AMP design methods typically optimize only predicted activity or toxicity, they do not directly evaluate how sequence changes affect evolutionary stability. By quantifying functional retention across mutational neighborhoods, this project addresses a documented gap in AMP design literature.

Algorithmically, the performance differences between GA, QIEA, and QIEA+RRI reflect more than incremental numerical gains; they indicate a fundamental change in how the peptide search space is explored and structured. QIEA+RRI not only achieved higher final composite fitness (0.8656 ± 0.0124) than both QIEA-only (0.8026 ± 0.0159) and GA (0.7318 ± 0.0217), but it also reached near-asymptotic performance substantially earlier in the optimization trajectory.

Reaching 90% of maximum fitness by generation 212—compared to 323 for GA—demonstrates accelerated organization of productive sequence regions. The larger area under the fitness curve further indicates that performance gains were sustained throughout the search process rather than confined to late-stage refinement. These patterns suggest that resilience-aware probabilistic updating reshapes the fitness landscape in a way that promotes both rapid ascent and stable plateauing in high-performing regions.

Population diversity dynamics clarify the mechanism underlying this behavior. The GA exhibited marked entropy collapse (1.118 ± 0.247 bits at generation 500), consistent with premature fixation on narrow sequence motifs. In contrast, QIEA-based methods retained substantially higher entropy (2.984 ± 0.074 bits for QIEA-only; 2.5671 ± 0.1925 bits for QIEA+RRI), approaching over half of the theoretical maximum (4.3219 bits). The slower entropy decay rates and higher mean pairwise Hamming distances indicate that probabilistic encoding preserves distributed sampling across amino acid positions, preventing early commitment to suboptimal motifs. Importantly, integrating RRI did not eliminate diversity; rather, it moderated it, suggesting guided exploration rather than unrestricted diffusion. This balance between controlled convergence and maintained dispersion is essential in high-dimensional peptide design spaces where early collapse can irreversibly limit accessible regions.

The multi-objective optimization results further demonstrate that resilience integration changes the qualitative structure of reachable solutions. QIEA+RRI produced 31 Pareto-optimal solutions compared to 14 for QIEA-only and 7 for GA, and achieved the largest hypervolume (0.059670), representing a 68.88% improvement over GA and 31.62% over QIEA-only. More notably, QIEA+RRI uniquely populated high-performance regions defined simultaneously by

efficacy ≥ 0.95 , toxicity ≤ 0.20 , and RRI ≥ 0.80 , with 22 such solutions compared to none for GA. This indicates that durability is not merely correlated with activity or toxicity but enables access to distinct regions of objective space that are otherwise unreachable under conventional fitness formulations.

Exploration efficiency metrics reinforce this interpretation. QIEA+RRI evaluated more unique sequences per generation (172 ± 9) and accumulated a larger total number of distinct sequences (34,870) compared to GA (21,430). Reduced variance in unique sequence counts further suggests stable exploration rather than erratic search behavior. The probabilistic representation therefore appears to expand sampling breadth without sacrificing convergence efficiency, supporting effective traversal of complex combinatorial landscapes.

Downstream analyses of candidate peptides demonstrate that algorithmic improvements translate into biologically coherent outputs. The top candidates exhibited high predicted efficacy (0.963 ± 0.011), low predicted toxicity (0.171 ± 0.018), and elevated RRI (0.842 ± 0.027) within realistic physicochemical constraints (length 18–22 residues, net charge +6 to +9). These properties align with established membrane-active AMP profiles, indicating that optimization did not generate unrealistic or biophysically implausible sequences.

Novelty assessment confirms that performance gains were not driven by rediscovery of known peptides. With a maximum BLASTp identity of 41.2% and a mean identity of $29.8\% \pm 6.4\%$, the optimized sequences occupy previously unexplored regions of sequence space while remaining functionally coherent. Consistent classification by external AMP predictors further reduces the likelihood of model-specific bias.

Structural modeling of the rank 1 candidate provides mechanistic context for the resilience objective. The predicted helix exhibits clear amphipathic segregation, with a

continuous cationic surface spanning one helical face and a complementary hydrophobic surface on the opposite side. Because positive charge is distributed across multiple residues rather than concentrated at a single position, disruption of any individual charged residue would be unlikely to abolish the overall electrostatic interface. This distributed surface architecture is consistent with the observed mutational robustness profile and supports the interpretation that RRI-selected peptides occupy structurally stable regions of sequence space rather than relying on isolated functional hotspots.

Notably, mutational scanning revealed that high-RRI peptides retained $\geq 90\%$ predicted activity in 71.90% of mutants, compared to 0.10% for low-RRI peptides. The substantial difference in mean fitness retention (0.9045 ± 0.0994 versus 0.5250 ± 0.1873 ; $p = 4.74802e-289$) demonstrates that resilience optimization corresponds to measurable stability under perturbation rather than superficial objective reweighting.

However, several sources of variability and uncertainty must be considered. First, predictive model error contributes to uncertainty in all downstream analyses. The efficacy and toxicity models, while achieving high validation accuracy, are trained on curated datasets that may not fully represent the complete biochemical diversity of antimicrobial peptides. Model bias toward sequences within the training feature distribution was observed during development, particularly for outlier peptides. Although mitigation steps were taken—such as restricting sequence ranges and adopting mutation-sensitive regression approaches—residual prediction error may influence RRI calculations and fitness evaluations.

Second, the RRI validation dataset is limited in size. The (6) dataset contains a controlled set of peptides evaluated under consistent laboratory conditions, which strengthens internal validity. However, the number of AMPs available for correlation analysis remains modest.

Resistance evolution is influenced by strain background, environmental conditions, and experimental duration. Although the use of median MIC fold change reduces sensitivity to extreme outliers, variability across replicate evolutionary lines introduces biological noise. Therefore, while the observed correlation supports the biological relevance of RRI, further validation with additional experimental datasets would strengthen confidence in the metric.

There are also conceptual limitations to interpreting RRI. The metric quantifies mutational robustness under single-point substitutions. While this approach aligns with established mutational landscape theory, real-world resistance may involve multi-step adaptation, regulatory changes, or horizontal gene transfer. RRI does not directly model these complex evolutionary pathways. Furthermore, structural robustness and physicochemical invariance are approximated using computational embeddings and feature variance. These proxies capture aspects of stability but do not replace detailed molecular dynamics simulations or in vivo testing.

Arguments both support and challenge the central hypothesis. In support, the statistically significant RRI correlation and improved Pareto dominance provide quantitative evidence that durability optimization changes the search trajectory and produces distinct candidate profiles. Additionally, in-silico mutational analyses show that high-RRI candidates maintain predicted activity across a broader range of substitutions compared to reference peptides. Against the hypothesis, it is possible that high RRI scores partly reflect inherent physicochemical conservatism rather than true evolutionary durability. Peptides with stable charge and hydrophobic profiles may appear robust computationally but still face resistance mechanisms in clinical contexts. This limitation highlights the need for experimental confirmation.

Despite these limitations, the results provide strong positive conclusions. First, the RRI metric demonstrates measurable predictive validity against independent biological data. Second,

incorporating RRI into evolutionary optimization changes algorithmic behavior in a measurable and beneficial way. Third, the QIEA framework consistently outperforms traditional GA methods in convergence stability and diversity maintenance. These findings align with prior reports that probabilistic evolutionary representations improve exploration efficiency in complex optimization problems. By combining this search mechanism with a durability-aware objective, the framework introduces a new design paradigm.

The broader significance of this work lies in reframing antimicrobial peptide discovery. Rather than treating resistance as a downstream clinical issue, this study integrates evolutionary resilience directly into the design objective. Current computational AMP tools emphasize immediate potency, which may accelerate the cycle of resistance emergence. The Q-PROTEUS framework demonstrates that durability can be quantified, optimized, and validated computationally before synthesis. This approach shifts antimicrobial development toward proactive resistance management.

Future work would include wet-lab validation of the produced top-ranked candidates, which is essential to confirm that predicted durability translates into clinically usable measurable biological resilience. This would require extended multi-generation laboratory evolution experiments conducted over another several months to directly measure resistance emergence dynamics. Nonetheless, within the scope of this study, the evidence supports the hypothesis that proactive optimization for evolutionary robustness produces computational AMP candidates that are fundamentally superior in predicted durability without sacrificing efficacy or safety.

In conclusion, this study provides both conceptual and quantitative support for incorporating evolutionary resilience into antimicrobial peptide design. The Resistance Resilience Index correlates with experimental resistance data, and the QIEA+RRI optimization

framework demonstrates superior algorithmic performance and candidate quality. While further experimental lab validation is required, the results indicate that designing for durability is not only feasible but advantageous.

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